Benyu Wang

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Interests

- Broad interest in **Theoretical Computer Science**.
- Study and research experience in **Graph Algorithms** / **Combinatorics**.

Education

University of Michigan **Ann Arbor, US**
PhD student at Computer Science and Engineering (CSE) Aug 2023 – Now
- Advisor: *Prof. Thatchaphol Saranurak*
- GPA: **4.00**/4.00

Tsinghua University (Yao Class) **Beijing, China**
B.Eng., Computer Science and Technology Aug 2019 – Jun 2023
- Advisor: *Prof. Ran Duan*
- GPA: **3.90**/4.00 & For Yao Class courses only: **3.98**/4.00

Selected Courses: (A+/A stand for 95-100 in 100-point values, A+ is the best grade)
· Mathematics for Computer Science (A+) & Mathematics for Artificial Intelligence (A+)
· Theory of Computation (A+) & Design and Analysis of Algorithms (A+)

Experiences

University of Michigan **Ann Arbor, US**
Undergraduate Research Intern (Visitor) Feb 2022 – Aug 2022

INSAIT **Sofia, Bulgaria**
Research Intern (Visitor) Sep 2024 – Dec 2024

Teaching

Coach of the UMich ICPC team 2025 – Now

Graduate Student Instructor, Introduction to Algorithms Winter 2025, University of Michigan
Undergraduate algorithm course instructed by Prof. Thatchaphol Saranurak.

Teaching Assistant, Theory of Computation Spring 2023, Tsinghua University
Undergraduate complexity theory course instructed by Prof. Ran Duan.

Teaching Assistant, Design and Analysis of Algorithms Fall 2022, Tsinghua University
Graduate-Level algorithm course instructed by Prof. Ran Duan.

Services

Conference Review: STOC, FOCS, SODA, ICALP, ITCS, ESA, COCOON.

Journal Review: Algorithmica, TALG.

Selected Awards and Scholarships

ICPC Astana World Finalist (Rank 2 in ICPC ECNA Regional) 2023-2024
Excellent Academic Scholarship & **Excellent Art** Scholarship of Tsinghua University 2022
Gold Medal in Chinese Mathematical Olympiad (CMO) (ranked 86th in China) 2018
Silver Medal in National Olympiad in Informatics (NOI) (ranked 90th in China) 2018

Publications

Tight Conditional Lower Bounds for Vertex Connectivity Problems
Zhiyi Huang, Yaowei Long, Thatchaphol Saranurak, Benyu Wang
∘ STOC 2023, arXiv:2212.00359.

Undirected 3-Fault Replacement Path in Nearly Cubic Time
Shucheng Chi, Ran Duan, Benyu Wang, Tianle Xie
∘ ICALP 2025, arXiv:2411.18312.

Near-Optimal Fault-Tolerant Strong Connectivity Preservers
Gary Hoppenworth, Thatchaphol Saranurak, Benyu Wang
∘ FOCS 2025, arXiv:2510.02562.

Connectivity Oracle Under Vertex Failures by Shortcutting Unbreakable Decomposition
Xizhe Li, Yaowei Long, David Pidugu, Thatchaphol Saranurak, Benyu Wang
∘ ICALP 2026, arXiv:2605.07168.

Strong Connectivity Oracle for Multiple Failures via Rooted Flow-Augmentation
Aditya Anand, Thatchaphol Saranurak, Benyu Wang
∘ SODA 2027

A Note on the Hardness of Partially Dynamic Max Flow
Thatchaphol Saranurak, Benyu Wang
∘ SOSA 2027

Directed Global Minimum Cut in Almost-Linear Time
Henry Fleischmann, Jason Li, Thatchaphol Saranurak, Benyu Wang
∘ Manuscript, arXiv:2610.10898.

Approximating Directed Minimum Cut and Arborescence Packing via Directed Expander Hierarchies
Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Benyu Wang
∘ Manuscript, arXiv:2512.05300.